

Project Requirements Document

Team: Moose Blazers

Project: The Aerospace Capstone

Team: Jordan Pringle, Melissa Anewalt, Peter Gaede, Scott Walstead

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Intro:

In our capstone project we will showcase the low-power, high-performance processing of NVidia's Jetson TK1 board utilizing the advantages of GPGPU parallel processing (CPU+GPU) over serial (CPU) processing. A GUI will allow a user to choose among various computer vision applications such as video compression/quality and motion tracking parameters. Finally the application will display performance data comparing serial and parallel versions of the same application.

Glossary of Terms:

-GPGPU: General-Purpose computing on Graphics Processing Units where a computer's GPU handles some of the tasks alongside the CPU.

-Computer Vision: A field of study focusing on the processing and analysis of images and video on a computer to understand the scenes represented in them.

-CUDA: A parallel processing programming model created by NVidia to run programs on their lines of graphics processors.

System Architecture Overview:

Our applications will run onboard the NVidia Jetson TK1 board. The user can interact with the application via a connected keyboard and monitor or via SSH remote login. The user will provide an input video file, and can select between different options of computer vision analysis on the input file. The output video stream will play in real-time as analysis overlaid on the input file.

Requirements: (to expand)

-As a developer, I can understand CUDA so that I can write both serial (CPU) and parallel (CPU+GPU) versions of computer vision applications.

- As a developer, I can understand OpenCV so that I can write both serial (CPU) and parallel (CPU+GPU) versions of computer vision applications.

-As an end user, I can run a serial (CPU) version of an application and I can view its performance metric.

-As an end user, I can run a parallel (GPU) version of an application and I can view its performance metric.

-As an end user, I can select in the application GUI (via a drop-down list) to run the serial (CPU) or parallel (CPU+GPU) version.

-As a continuity checking application, the outputs of the serial/parallel versions of the same application are checked to ensure both give the same results.

- As a motion tracking application, the activity of tracked objects can be measured so that the resolution and compression of a video stream can be adjusted accordingly.
- As a motion tracking application, the resolution and data sampling rate will increase when activity in the scene passes above a threshold in order to display interesting information in detail.
- As a motion tracking application, the resolution and data sampling rate will decrease when activity in the scene passes below a threshold in order to conserve power when nothing interesting is happening.
- As a performance metric, the performance of both versions will be displayed graphically and textually.

Test Cases:

<https://github.com/scotty914/GPGPU-Aerospace/>

- Be able to write a parallelized function using CUDA.
- Be able to write a parallelized function using OpenCV.
- Be able to select serial processing or parallel processing from a drop down menu
- Be able to select a video file and load it into the GUI
- Is there a dropdown menu? Does that menu have two options, CPU and GPU? Does selecting these options call their corresponding functions?
- Does the test application that compares the outputs of the CPU and GPU applications return with 0? If they do not return 0, return error message.
- Does the application display the speed of a tracked object? Is the displayed speed accurate?
- When the activity in the scene is above a set threshold, does the resolution increase to a high set value?
- When the activity in the scene is below a set threshold, does the resolution decrease to a low set value?
- Does the conclusion of an application bring up a performance statistics page? Are the performance statistics relevant to the application that just ran?

Technologies:

Hardware:

- NVIDIA Jetson TK1 for running code

Software/Libraries:

- OpenCV for reading images and videos
- CUDA for parallelizing code on the GPU
- Thrust?